

Ubiquitination-activated TAB-TAK1-IKK-NF- κ B axis modulates gene expression for cell survival in the lysosomal damage response

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This study presents the **important** finding that lysosomal damage triggers inflammatory signaling through ubiquitination and the TAB-TAK1-IKK-NF- κ B axis. The data obtained from the unbiased transcriptomic and proteomic analyses are **convincing** and provide invaluable information to the field. Although further experiments will be required to clarify how TAB2/3 are recruited after various types of lysosome damage, this work will be of interest to researchers in the fields of organelle biology and inflammation.

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Abstract

The lysosomal damage response is important for the maintenance of cellular homeostasis. Although the mechanisms underlying the repair and autophagic elimination of damaged lysosomes have been elucidated, the early signal transduction pathways and genes induced in response to lysosomal damage remain elusive. We performed transcriptome and proteome analyses and found that the TAB-TAK1-IKK-NF- κ B axis is activated by K63-linked ubiquitin chains that accumulate on damaged lysosomes. This activates the expression of various transcription factors and cytokines that promote anti-apoptosis and intercellular signaling. The findings highlight the crucial role of ubiquitin-regulated signal transduction and gene expression in cell survival and cell-cell communication in response to lysosomal damage. The results suggest that the ubiquitin system is not only involved in the removal of damaged lysosomes by lysophagy, but also functions in the activation of cellular signaling for cell survival.

Introduction

Lysosomes are the primary degradative organelles and play a pivotal role in maintaining cellular and tissue homeostasis^{1,2}. Lysosomal dysfunction is associated with various diseases, underscoring the crucial role of lysosomal integrity and its quality control in human health^{3–5}. Lysosomes are susceptible to damage by many sources, both internal and external to the cell, which results in the permeabilization and rupture of lysosomal membranes. Crystals, amyloids, pathogens, lipids, lysosomal toxic drugs, and reactive oxygen species (ROS) are among the stressors that can damage lysosomal membranes^{6–10}. The release of enzymes and protons from the lumen of damaged lysosomes induces oxidative stress, inflammation, and cell death, which is deleterious to cells and tissues^{10,11}. The accumulation of damaged lysosomes is thus an important factor involved in the development of aging-related diseases and other conditions such as crystalline nephropathy, neurodegenerative, and infectious diseases^{3–5,12}.

To overcome lysosomal damage and restore lysosomal functionality, cells have developed a set of response pathways that is collectively referred to as the lysosomal damage response^{6,13}. It has been reported that signal mediators, such as AMP-activated protein kinase (AMPK) and mammalian target of rapamycin (mTOR), exert the signal transduction in the lysosomal damage response^{14,15}. The quality control mechanisms for damaged lysosomes include repair, elimination, and regeneration. Lysosomal membrane damage triggers the recruitment of the endosomal sorting complex required for transport (ESCRT) to repair the damaged membrane^{13,16}. Current studies have shown that endoplasmic reticulum (ER)-lysosome lipid transfer is crucial for lysosomal membrane repair^{17,18}. Furthermore, stress granules (SGs) have been demonstrated to play a pivotal role in the repair process¹⁹. When the damage is extensive and repair is not feasible, the damaged lysosome undergoes autophagic elimination in a process known as lysophagy^{13,20}. In lysophagy, lysosomal membrane permeabilization leads to extensive ubiquitination of exposed proteins. This triggers the recruitment of ubiquitin-binding autophagy receptors, which activates autophagy to sequester and degrade damaged lysosomes^{13,21}. Protein ubiquitination is an important form of post-translational modification that is comparable to protein phosphorylation. Protein ubiquitination in lysophagy occurs through the formation of Lys-63-linked ubiquitin chains (K63 ubiquitin chains)²². Substrate ubiquitination in damaged lysosomes is catalyzed by the E2 ubiquitin-conjugating enzyme UBE2QL1 and several ubiquitin E3 ligases, including TRIM16, CUL1–SKP1–FBXO27, and CUL4A–DDB1–WDFY1^{23–26}.

Transcription factor EB (TFEB), a master regulator of lysosome- and autophagy-related gene expression, is activated during the repair and elimination of damaged lysosomes, which results in the regeneration of lysosomes as part of the lysosomal damage response^{27–29}. Under unstimulated conditions, TFEB is phosphorylated by lysosome-associated mTORC1 and binds to 14–3–3 proteins, resulting in the cytoplasmic retention of TFEB³⁰. However, lysosomal permeabilization causes the dissociation of mTORC1 from lysosomes, which leads to the dephosphorylation of TFEB and its translocation to the nucleus to activate the expression of lysosome- and autophagy-related genes^{3,14,30}.

The elimination of damaged lysosomes through lysophagy requires a period of more than half a day, during which the cell is vulnerable³¹. It is therefore assumed that there are cellular mechanisms that cooperate with lysophagy to maintain cellular health in the lysosomal damage response. In previous work, we demonstrated that K63 ubiquitin chains, which accumulate on dysfunctional endosomes, induce cytokine production by modulating gene expression in a TGF-beta-activated kinase 1 (TAK1) and TAK1-binding protein (TAB) dependent manner^{32,33}. K63 ubiquitin chains also accumulate in damaged lysosomes, which serves as an initiating signal for lysophagy. This prompted us to investigate whether gene expression is induced by ubiquitin-

mediated activation of the TAB–TAK1 pathway in the lysosomal damage response, as observed in the endosomal stress response. The transcriptome of cells exposed to lysosomal damaging agents for 30 min has been described³⁴; however, genome-wide gene expression patterns associated with the lysosomal damage response remain unclear. In this study, we performed transcriptome and proteome analyses to comprehensively examine the regulation of gene expression in the lysosomal damage response and to identify the underlying signal transduction pathways. We found that accumulation of K63 ubiquitin chains in damaged lysosomes activates the TAB–TAK1–IKK–NF-κB pathway, which in turn induces the expression of various transcription factors and cytokines. We elucidated the critical role of the TAB–TAK1–IKK–NF-κB pathway in cell survival and intercellular signal transduction. Collectively, the findings suggest the existence of a universal response mechanism against the intracellular accumulation of K63 ubiquitin chains.

Results

Lysosomal damage has a global impact on the transcriptome and proteome

To comprehensively investigate the changes in gene expression caused by lysosomal damage, we performed transcriptome analysis and compared mRNA expression between cells treated with L-Leucyl–L-Leucine methyl ester (LLOMe), a well-characterized inducer of lysosomal damage³⁵, and control cells. We used RPE-1 cells, which are commonly used in aging research. Transcriptome data revealed substantial alterations in gene expression in response to lysosomal damage. We identified >1,000 genes (946 upregulated and 164 downregulated genes from >13,000 transcripts with Log₂ fold change (FC) >|1| and *P* <0.05) that exhibited significant changes at 2 h after LLOMe treatment (**Fig. 1A**). Proteome analysis was performed to determine the correlation between proteomic changes and the observed alterations in the transcriptome. The analysis identified 86 upregulated and 60 downregulated proteins (of >10,000 proteins with Log₂ FC >|1| and *P* <0.05) (**Fig. 1B**). There was a modest correlation between the significant upregulation at the RNA level and that at the protein level, with a correlation coefficient (*R*) of 0.5899 from 302 data points (**Fig. 1C** and **Fig. S1A**). The representative targets interleukin-1 β (IL1β), IL6, interferon regulatory factor 1 (IRF1), and the proto-oncogenes c-Jun (JUN) and c-Fos (FOS) were upregulated at both the RNA and protein levels (**Fig. 1C** and **S1A, B**). This indicates that alterations in the transcriptome were, at least in part, responsible for those observed in the proteome. Genes showing significant upregulation in the transcriptome and proteome analyses were subjected to gene enrichment analysis using the Gene Ontology Molecular Function (GOMF) reference dataset (**Fig. 1D, E**). The results showed that these genes were primarily enriched in transcription factors and cytokines (labeled in red and blue, respectively, in **Fig. 1D**). Analysis using a Molecular Signatures Database (MSigDB) hallmark gene set³⁶, a phenotypic reference dataset, indicated that genes upregulated at both the RNA and protein levels were involved in biological regulatory processes including the inflammatory response and apoptosis (**Fig. 1E**). It is noteworthy that the transcription factor prediction using the DoRothEA regulon^{37,38} identified the p65 (RELA) and p105 (NFKB1) subunits of the NF-κB family as the primary transcription factors regulating gene expression in the lysosomal damage response (**Fig. 1F**). This suggests that NF-κB activation is involved in the regulation of gene expression in the lysosomal damage response. The upregulation of IL1β, IL6, IRF1, JUN, and FOS at the protein level was confirmed by immunoblotting (**Fig. 1G**).

Taken together, the results suggest that in the lysosomal damage response, the proteome is remodeled according to alterations in the transcriptome; this remodeling is mediated primarily through NF-κB and involves many transcription factors and cytokines, which may potentially lead to the regulation of inflammation and apoptosis (**Fig. 1H**).

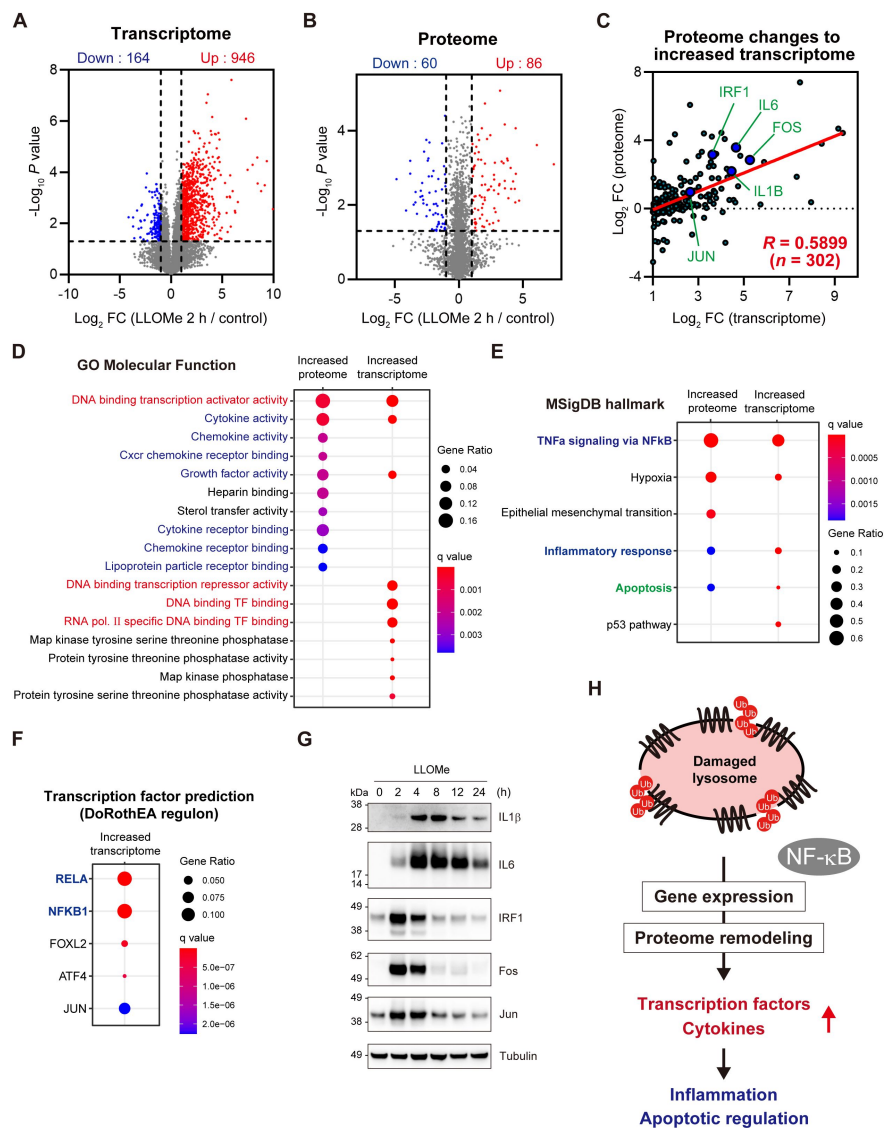


Figure 1

Lysosomal damage has a global impact on the transcriptome and proteome

(A, B) The mean Log₂ fold change (FC, LLOMe 2 h/control) and -Log₁₀ P value of the transcriptome (A) and proteome (B) in RPE-1 cells are indicated on the x and y axes, respectively. Genes significantly upregulated or downregulated are labeled in red and blue, respectively.

(C) For genes showing significant upregulation in the transcriptome (Log₂ FC > 1 and $P < 0.05$), the correlation between the mean Log₂ FC of the transcriptome (from A, x axis) and the proteome (from B, y axis) is shown with a coefficient of correlation ($R = 0.5899$, $n = 302$).

(D, E) The bubble plots show the outcomes of gene enrichment analyses based on gene ontology molecular function (GOMF) (D) and MSigDB hallmark gene sets (E) for genes upregulated at the protein and RNA levels (top ten and five categories for GOMF and MSigDB hallmark gene sets, respectively). The color and size of the bubbles indicate the q value and gene ratio, respectively. The categories related to transcription, cytokine/growth factor, and apoptosis are labeled in red, blue, and green, respectively.

(F) The bubble plot illustrates the DoRothEA regulon-based prediction of transcription factors responsible for the induction of genes upregulated in cells treated with LLOMe (top five transcription factors). The color and size of the bubbles indicate the q value and gene ratio, respectively. NF-κB components are labeled in blue.

(G) RPE-1 cells were treated with LLOMe for 2 h and then washed. The total incubation times are indicated. Total cell lysates were subjected to immunoblotting with the indicated antibodies.

(H) Schematic model of the cellular response to lysosomal damage.

Lysosomal damage activates TAK1 in a ubiquitin- and TAB-dependent manner

Lysosomal membrane damage exposes glycoproteins, such as LAMP2. Cytosolic galectin proteins, including Galectin-3 (Gal-3), are recruited to the lysosomal membrane by recognizing beta-galactoside-containing glycans on exposed proteins. These exposed lysosomal proteins are extensively ubiquitinated by ubiquitin ligases, resulting in the accumulation of ubiquitinated proteins within damaged lysosomes^{25,26,39}. An endosomal stress response mechanism was recently described whereby K63 ubiquitin chains accumulate on dysfunctional endosomes, activating TAK1 in a TAB-dependent manner³². TAB2 and TAB3 (a paralog of TAB2) function as ubiquitin decoders that specifically recognize K63 ubiquitin chains via the C-terminal Npl4 zinc finger (NZF) domain^{40,41}. The finding that TAB binding to unanchored K63 ubiquitin chains is sufficient to activate TAK1 in vitro⁴² suggests that the conjugation of K63 ubiquitin chains to bulky substrates on endosomes is sufficient to facilitate TAB-dependent TAK1 activation. This prompted us to investigate whether the accumulation of K63 ubiquitin chains in other regions of the cytoplasm also results in TAK1 activation. TAK1 controls the function of the p62/sequestosome, an autophagy receptor, and TAB2 undergoes TRIM38-mediated lysosomal degradation in response to lysosomal damage^{43,44}. However, whether TAK1 is activated in a TAB-dependent manner and the role of the TAB–TAK1 signaling pathway in the lysosomal damage response remain unanswered questions.

To determine whether the TAB–TAK1 signaling pathway is activated in response to lysosomal damage, we initially examined the subcellular localization of TAB2 and TAK1. The recruitment of Gal-3 and K63 ubiquitin chains to damaged lysosomes was confirmed in RPE-1 cells (**Fig. S2A–B**). The co-localization of TAB2 and TAK1 with K63 ubiquitin chains at damaged lysosomes was observed 5 min after LLOMe treatment (**Fig. 2A**). The signals of LAMP1, a lysosomal marker protein, were enhanced, and LAMP1-positive puncta were increased in size in RPE-1 cells treated with LLOMe. It is plausible that this phenomenon reflects the morphological alterations in lysosomes under our experimental conditions, possibly due to lysosomal membrane damage (**Fig. 2A**, **S2A**). Immunoblotting analysis showed that the accumulation of K63 ubiquitin chains in the cell was accompanied by an increase in active phosphorylated TAK1, which was first detected 5 min after LLOMe treatment (**Fig. 2B**). TFEB isolated from cells treated with LLOMe exhibited a downshift on the gel, which is a characteristic of TFEB activation⁴⁵. This indicates that a lysosomal damage response was activated under these experimental conditions (**Fig. 2B**). TAK1 activation was abolished by the E1 ubiquitin-activating enzyme inhibitor TAK-243 and depletion of TAB2 and TAB3 (**Fig. 2C, D**). These results suggest that TAK1 is activated in a ubiquitin- and TAB-dependent manner in the lysosomal damage response. To further explore the potential involvement of K63 ubiquitination in TAK1 activation, we performed an add-back experiment using siRNA-resistant TAB2 WT and mutants incapable of binding to K63 ubiquitin chains, dNZF and E685A. The NZF domain of TAB2 is well-characterized to specifically associate with K63 ubiquitin chains⁴⁰. We investigated whether the add-back of TAB2 mutants rescues the activation of TAK1 in TAB2-depleted cells (**Fig. 2E**). TAB2 WT, but not dNZF and E685A, restored TAK1 activation in response to lysosomal damage, suggesting that the specific interaction of TAB proteins and K63 ubiquitin chains is a key mechanism to activate TAK1. Unexpectedly, the treatment of an E1 inhibitor TAK-243 that abolished the lysosomal accumulation of K63 ubiquitin chains did not inhibit the recruitment of TAB2 to damaged lysosomes, suggesting that the recruitment of TAB proteins to damaged lysosomes is independent of the association with K63 ubiquitin chains (**Fig. S2B**). Collectively, it is postulated that TAB proteins require the interaction with K63 ubiquitin chains for TAK1 activation, but not for recruitment to damaged lysosomes. We next examined whether other lysosomal damage inducers, Glycyl-L-phenylalanine 2-naphthylamide (GPN) and DC661, also activate TAK1. It has been shown that GPN and DC661 induce lysosomal membrane permeabilization through hyperosmotic stress and lipid peroxidation in lysosomes, respectively^{46,47}. In cells treated with GPN and DC661, K63 ubiquitin chains

were accumulated, possibly due to a defect in the endo-lysosome machinery (Fig. S2C, D). However, the lysosomal accumulation of Gal-3 was significantly lower than that observed in cells treated with LLOMe, suggesting that GPN and DC661 did not induce severe lysosomal membrane permeabilization (Fig. S2C, D). Under these conditions, the phosphorylation of TAK1 was not observed (Fig. S2E). These findings indicate that the exposure of K63 ubiquitin chains within the cytosol following severe lysosomal membrane permeabilization functions as an initial trigger for TAB-mediated TAK1 activation in the lysosomal damage response.

The TAB–TAK1 pathway plays a pivotal role in the induction of cytokines and transcription factors in the lysosomal damage response

To gain further insight into the involvement of the TAB–TAK1 pathway in the regulation of gene expression associated with the lysosomal damage response, an additional transcriptome analysis was performed on TAB- and TAK1-depleted cells treated with LLOMe for 2 h. The genes upregulated in response to lysosomal damage were classified into six distinct clusters within the dataset (Fig. 3A). Clusters 4 and 5 comprised genes that are expressed in a TAB- and TAK1-dependent manner (Fig. 3A). Most of the genes upregulated in a TAB- and TAK1-dependent manner showed overlapping expression patterns (Fig. S3A, B); these genes were enriched for cytokines and growth factors, and their functions were linked to inflammatory responses and apoptosis, as shown in the gene enrichment analysis (Fig. S3C, D). Transcription factor prediction based on the DoRothEA regulon indicated that genes in cluster 4, whose expression was inhibited by depletion of TAB and TAK1, were targets of the NF- κ B family, including RelA, NFKB1, and c-Rel (REL) (Fig. 3B). The NF- κ B family members were identified as the primary transcription factors regulating gene expression in the lysosomal damage response (Fig. 1F). This indicates that the TAB–TAK1 pathway serves as a primary axis for the activation of NF- κ B in the lysosomal damage response, thereby inducing target gene expression. Cytokines and growth factors were enriched in cluster 4, and transcription factors were enriched in cluster 5, in which gene expression was partially suppressed (Fig. 3C, E). Regulators of the inflammatory response and apoptotic functions were commonly enriched in clusters 4 and 5 (Fig. 3D). The results of immunoblotting showed that the protein expression of IL1 β , IL6, IRF1, homeobox protein Nkx-3.1 (NKX3.1), and JUN was consistent with the TAB- and TAK1-mediated gene expression. Conversely, depletion of TAB and TAK1 did not affect the expression of FOS at both the mRNA and protein levels (Fig. 3F and S3E). Furthermore, the downshift of TFEB, which indicates the activation of TFEB, was not affected by depletion of TAB and TAK1 (Fig. 3F). This suggests that TFEB activation is independent of the TAB–TAK1 pathway in the lysosomal damage response.

In conclusion, transcriptome analysis in TAB- and TAK1-depleted cells indicates that the TAB–TAK1 pathway induces the expression of cytokines and transcription factors, potentially via NF- κ B, in the lysosomal damage response, thereby regulating inflammatory response and apoptosis.

The K63 Ub–TAB–TAK1–IKK–NF- κ B pathway is activated in the lysosomal damage response

To elucidate the downstream cascade of the TAB–TAK1 signaling pathway in the early response to lysosomal damage, we performed proteome and phosphoproteome analyses using the TAK1 inhibitor HS-276 (Fig. 4A, S4A). The phosphoproteome analysis showed that the phosphorylation levels of stress-induced mitogen-activated protein kinases (MAPKs), specifically p38 α (MAPK14, pT180/pY182) and JNK3 (MAPK10, pY223), as well as MK2 (MAPKAPK2, pT334), a downstream target of MAPKs, increased in response to lysosomal damage, and this increase was dependent on TAK1 (Fig. S4B, C). Phosphorylation of T180/Y182 in p38 α and Y223 in JNK3 activates the kinase function^{48,49}. The TAK1-dependent phosphorylation of Y223 in JNK3 in the lysosomal damage response was consistent with a previous report that lysosome rupture

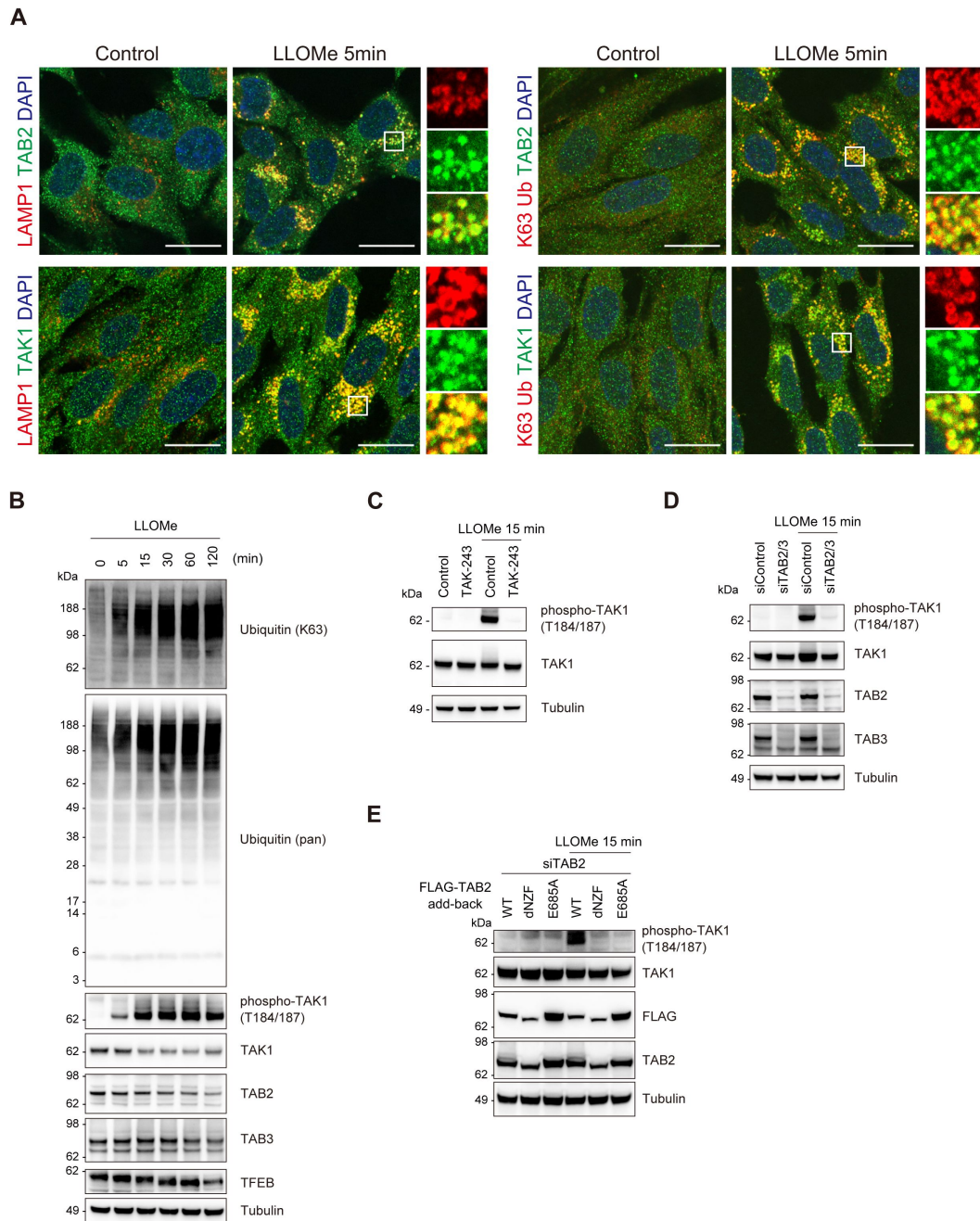


Figure 2

Lysosomal damage activates TAK1 in a ubiquitin- and TAB-dependent manner

(A) RPE-1 cells treated with LLOMe for 5 min were immunostained with the indicated antibodies and DAPI. Scale bar, 20 μ m.

(B) Total cell lysates from RPE-1 cells treated with LLOMe for the indicated times were subjected to immunoblotting with the indicated antibodies.

(C) Total cell lysates from RPE-1 cells pre-treated with TAK-243 for 15 min and treated with LLOMe for 15 min were subjected to immunoblotting with the indicated antibodies.

(D) Total cell lysates from RPE-1 cells transfected with the indicated siRNAs and treated with LLOMe for 15 min were subjected to immunoblotting with the indicated antibodies.

(E) HeLa cells stably expressing FLAG-TAB2 WT, dNZF, and E685A were transfected with the indicated siRNAs and treated with LLOMe for 15 min. Total cell lysates were subjected to immunoblotting with the indicated antibodies.

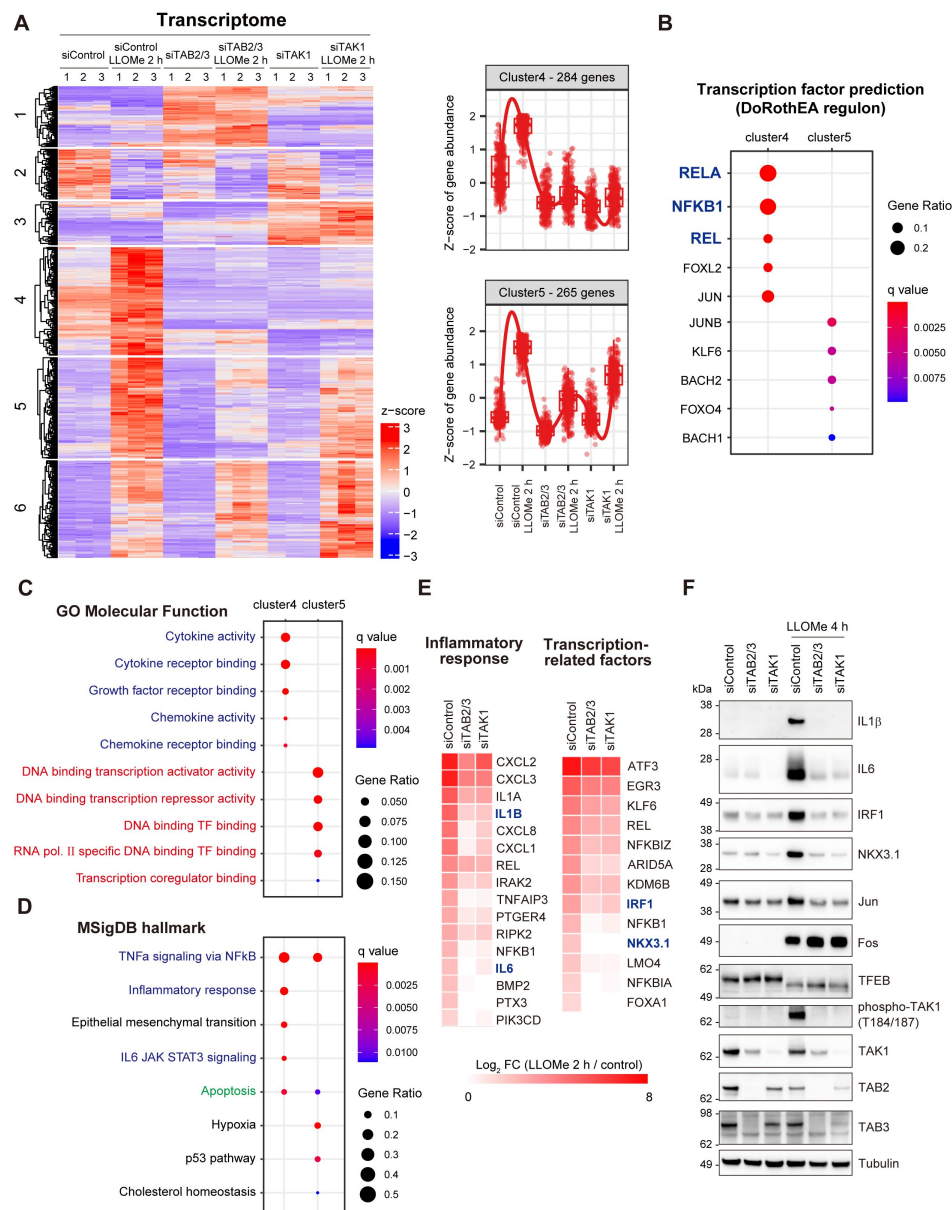


Figure 3

The TAB-TAK1 pathway activates cytokines and transcription factors in response to lysosomal damage

(A) Heatmap showing the changes in the transcriptome of each sample. Clusters 4 and 5 include the genes upregulated by LLOMe in a TAB- and TAK1-dependent manner.

(B) The bubble plot shows the DoRothEA regulon-based prediction of transcription factors responsible for the induction of genes assigned to clusters 4 and 5 in (A) (top five transcription factors). The color and size of bubbles indicate *q* value and gene ratio, respectively. NF-κB components are labeled in blue.

(C, D) The bubble plots show the outcomes of gene enrichment analyses based on GOMF (C) and MSigDB hallmark gene sets (D) for genes assigned to clusters 4 and 5 in (A) (top five categories). The color and size of the bubbles indicate the *q* value and gene ratio, respectively. The categories related to transcription, cytokine/growth factor, and apoptosis are labeled in red, blue, and green, respectively.

(E) Heatmaps showing the genes upregulated in a TAB- and TAK1-dependent manner within the categories of inflammatory response (left) and transcription-related factors (right). The color intensity indicates the Log₂ FC (LLOMe 2 h/control).

(F) Total cell lysates from RPE-1 cells transfected with the indicated siRNAs, treated with LLOMe for 2 h, and washed for 2 h were subjected to immunoblotting with the indicated antibodies.

activates the TAK1–JNK pathway in THP-1 cells⁵⁰. Additionally, T334 of MK2 is a primary site of phosphorylation by MAPKs⁵¹. These findings suggest that lysosomal damage induces TAK1-dependent activation of MAPKs. Proteome analysis showed that the inhibitor of NF- κ B α (I κ B α) undergoes TAK1-dependent degradation 30 min after LLOMe treatment (**Fig. 4A, B**). The inhibitor of NF- κ B kinase (IKK) complex, which comprises IKK α , β , and γ , functions downstream of TAK1 and phosphorylates I κ B α , thereby inducing its degradation⁵². Similarly, in response to ubiquitin-mediated endosomal stress, the NF- κ B pathway is activated in an IKK-dependent manner³². The degradation of I κ B α is a hallmark phenomenon upstream of NF- κ B activation⁵³, which is primarily triggered by IKK-mediated phosphorylation. This leads to the hypothesis that the IKK complex activated by the TAB–TAK1 pathway promotes the phosphorylation of I κ B α , which leads to its degradation and NF- κ B activation in the lysosomal damage response.

We therefore sought to characterize the kinase cascade that is activated in the lysosomal damage response. We confirmed the phosphorylation and subsequent degradation of I κ B α (**Fig. 4C**). We also found that MAPKs and IKK were activated concomitant with a notable phosphorylation of IKK, JNK, and p38, and a relatively minimal phosphorylation of ERK (**Fig. 4C**). The increase in the phosphorylation levels of IKK, JNK, p38, and I κ B α occurred in a ubiquitin-dependent manner, as the induction was abolished by TAK-243 (**Fig. 4D**). Depletion of TAB, TAK1, and IKK suppressed I κ B α phosphorylation, and the phosphorylation of JNK and p38 was TAB- and TAK1-dependent (**Fig. 4E, F**). These findings indicate that the phosphorylation of I κ B α is induced in a TAB–TAK1–IKK pathway-dependent manner. Additionally, it can be postulated that MAPKs, including JNK and p38, are activated by the TAB–TAK1 pathway.

TANK-binding kinase 1 (TBK1), a putative NF- κ B regulator, has been reported to be activated in the lysosomal damage response²¹. We examined whether TBK1 coordinately functions with the TAB–TAK1 pathway to facilitate NF- κ B activation. Consistent with the previous report, TBK1 was activated in response to LLOMe treatment (**Fig. S4D**). Depletion of TAB and TAK1 led to a slight reduction in the activation of TBK1 (**Fig. S4E**). The TBK1 inhibitor BX-795 did not affect TAK1 activation but abolished the phosphorylation of IKK and I κ B α (**Fig. S4F**). With regard to the ubiquitination in the lysosomal damage response, the linear ubiquitin chain assembly complex (LUBAC)-mediated M1 ubiquitination has been implicated in NF- κ B activation⁵⁴. To explore the functional link between K63 and M1 ubiquitination, we examined the effects of LUBAC depletion on the TAB–TAK1 pathway and the subsequent NF- κ B activation. Depletion of RNF31 (also known as HOIP), a component of LUBAC, exhibited no or minimal effects on TAK1 activation, but abolished the phosphorylation of IKK and I κ B α (**Fig. S4G**). These findings suggest that TBK1 and LUBAC are indispensable for NF- κ B activation in the lysosomal damage response.

The IKK complex, which is activated by the TAB–TAK1 pathway in response to lysosomal damage, has been shown to induce the phosphorylation and degradation of I κ B α , potentially leading to the activation of NF- κ B (**Fig. 4G**). Furthermore, TBK1 and LUBAC were required for NF- κ B activation, but not for TAK1 activation, indicating that TBK1 and LUBAC function downstream of or parallel to the TAB–TAK1 pathway.

The K63 Ub–TAB–TAK1–NF- κ B pathway regulates the expression of cytokines and transcription factors essential for cell survival and intercellular signaling

IL1 β , IL6, IRF1, and NKX3.1 expression was associated with the lysosomal damage response and dependent on the TAB–TAK1 pathway (**Fig. 3E, F, and S3E**). The expression of these genes was suppressed by treatment with TAK-243 and HS-276 (**Fig. 5A, B**). Depletion of IKKs suppressed target gene expression (**Fig. 5C**). These findings demonstrate that the expression of these genes is induced by the ubiquitin-dependent activation of the TAB–TAK1–IKK pathway in the lysosomal damage response.



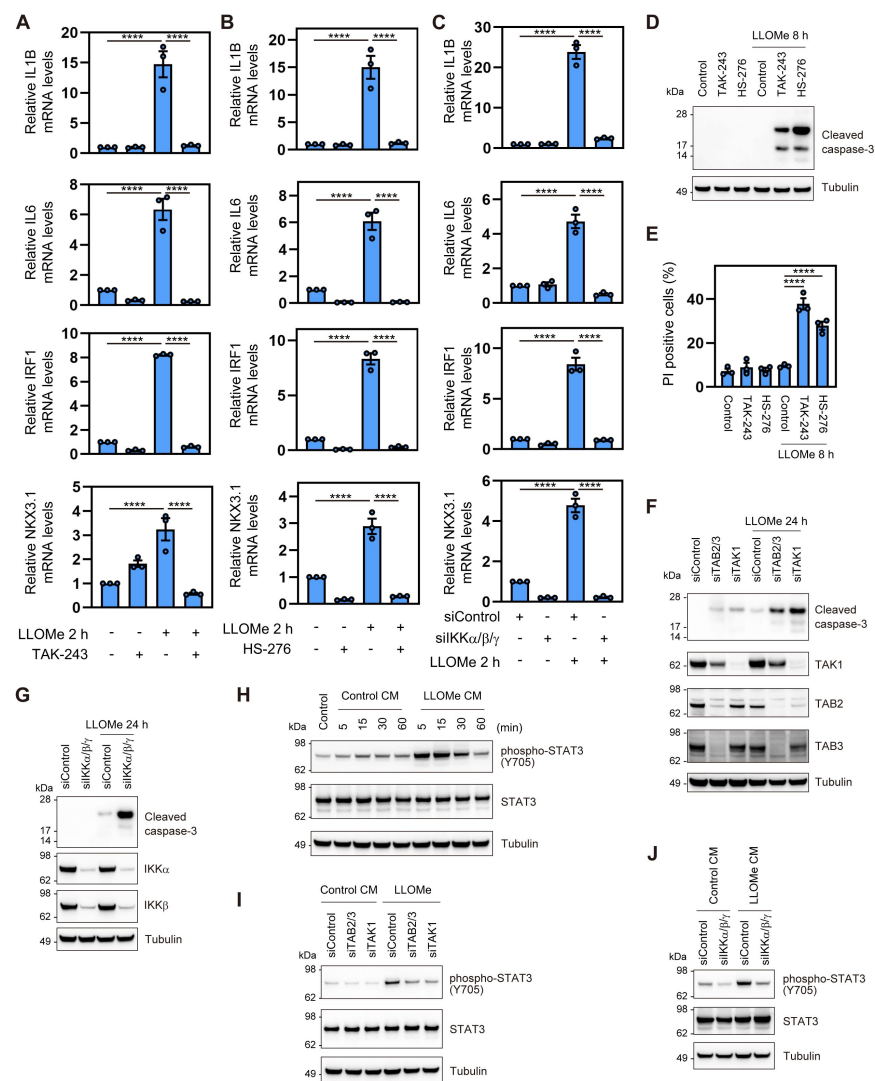


Figure 5

The K63 Ub-TAB-TAK1-IKK-NF-κB pathway promotes cell survival and intercellular signaling

(A, B) Total RNA from RPE-1 cells pre-treated with TAK-243 (A) and HS-276 (B) for 15 min and treated with LLOMe for 2 h was analyzed by RT-qPCR. Target mRNA levels were normalized to GAPDH mRNA levels; the expression levels in control cells were set to 1. The individual values, mean, and standard error of the mean (SEM) of relative mRNA levels are presented. The mean $\pm$ SEM values were calculated from three biological replicates. **** $P < 0.0001$ (one-way ANOVA with Dunnett's test).

(C) Total RNA from RPE-1 cells transfected with the indicated siRNAs and treated with LLOMe for 2 h was analyzed by RT-qPCR. Target mRNA levels were normalized to GAPDH mRNA levels; the expression levels in cells treated with control siRNA were set to 1. The individual values, mean, and SEM of relative mRNA levels are presented. The mean $\pm$ SEM values were calculated from three biological replicates. **** $P < 0.0001$ (one-way ANOVA with Dunnett's test).

(D, E) HeLa cells were pre-treated with TAK-243 or HS-276 for 15 min, treated with LLOMe for 2 h, and washed for 6 h. Total cell lysates were subjected to immunoblotting with the indicated antibodies (D). Cells were stained with propidium iodide (PI), and the fractions of PI-positive cells were assessed using the flow cytometer. Error bars indicate SD ($n = 3$). **** $P < 0.0001$ (one-way ANOVA with Dunnett's test) (E).

(F, G) Total cell lysates from HeLa cells transfected with the indicated siRNAs, treated with LLOMe for 2 h, and washed for 22 h were subjected to immunoblotting with the indicated antibodies.

(H) RPE-1 cells were stimulated for the indicated times with conditioned media (CM) from RPE-1 cells treated with LLOMe for 30 min and washed for 7.5 h. Total cell lysates were subjected to immunoblotting with the indicated antibodies.

(I, J) RPE-1 cells were stimulated for 15 min with CM from RPE-1 cells transfected with the indicated siRNAs, treated with LLOMe for 30 min, and washed for 7.5 h. Total cell lysates were subjected to immunoblotting with the indicated antibodies.

Gene enrichment data based on the results of transcriptome and proteome analyses indicated that TAB–TAK1 pathway-dependent gene expression plays a role in the regulation of apoptosis and the inflammatory response (**Fig. 3D** [↗](#) and **S3D**). We thus examined the impact of the K63 Ub–TAB–TAK1–IKK–NF- κ B pathway on these cellular functions. Lysosomal damage for 8 h did not induce apoptosis; however, inhibition of ubiquitination by TAK-243 dramatically induced apoptosis in HeLa and RPE-1 cells (**Fig. 5D, E, S5A** [↗](#)). This indicates that ubiquitination is essential for cell survival in the lysosomal damage response. Similarly, inhibition of TAK1 by HS-276 and depletion of TAB, TAK1, and IKKs following lysosomal damage for 24 h promoted apoptosis in HeLa cells (**Fig. 5D–G** [↗](#)). TAK1 inhibition induced apoptosis in RPE-1 cells to a lesser extent compared to HeLa cells (**Fig. 5D, E, S5A** [↗](#)). In addition, the inhibition of JNK by JNK-IN-8 promoted apoptosis following lysosomal damage for 12 h, which is a later time point than TAK-243 and HS-276 (**Fig. S5B** [↗](#)). These findings demonstrate that the K63 Ub–TAB–TAK1 pathway coordinately exerts anti-apoptotic effects through the downstream effectors such as IKK and JNK in response to lysosomal damage, thereby promoting cell survival.

We further investigated the potential autocrine/paracrine-like effects of cytokines such as IL1 β and IL6, which are expected to be secreted outside the cells due to their upregulation in the lysosomal damage response. IL6 activates the JAK–STAT pathway, which is accompanied by the phosphorylation of STAT3⁵⁵ [↗](#). Stimulation of untreated cells with conditioned media (CM) from lysosome-damaged cells resulted in the phosphorylation of STAT3, which indicates kinase activation (**Fig. 5H** [↗](#)). These results suggest that the lysosomal damage response activates intercellular signaling involving the JAK–STAT pathway. Furthermore, the capacity to activate this intercellular signaling was decreased by depletion of TAB, TAK1, and IKKs (**Fig. 5I, J** [↗](#)).

Collectively, these findings suggest that the K63 Ub–TAB–TAK1 pathway plays a pivotal role in cell survival and intercellular signaling in response to lysosomal damage.

Discussion

In this study, comprehensive analyses of the transcriptome and proteome demonstrated that lysosomal damage causes substantial alterations in gene expression, leading to proteome remodeling. Furthermore, we identified the signal transduction pathway mediating this response (**Fig. 6** [↗](#)). The accumulation of K63 ubiquitin chains in damaged lysosomes activates the TAB–TAK1–IKK–NF- κ B pathway, which induces the expression of various transcription factors and cytokines. The present findings suggest that the ubiquitin-dependent signaling pathway plays a regulatory role in cell survival and intercellular signaling in the lysosomal damage response.

Prior to this study, our understanding of the lysosomal damage response was largely limited to the processes of repair and elimination of damaged lysosomes, as well as the activation and function of TFEB in lysosomal regeneration. Recent findings highlighting the role of stress granule formation in the lysosomal damage response suggest that additional pathways are involved in this process beyond what was previously anticipated³⁴ [↗](#). Nevertheless, the complete picture remains to be elucidated. This study provides new information on the regulation of gene expression and signal transduction in the lysosomal damage response that offers a novel perspective for future research.

Aberrations in the lysosomal damage response are relevant to the pathogenesis of various age-related diseases, including neurodegenerative diseases^{3,56} [↗](#). In the context of pathology, inflammation, which is associated with numerous diseases, is linked to lysosomes. The inflammatory response is not only regulated by lysosomal function, but also triggered by lysosomal damage, and observed in lysosomal storage diseases^{57,58} [↗](#). The NLRP3 inflammasome induces an inflammatory response upon lysosomal damage by increasing cytokine processing at the protein level^{58,59} [↗](#). The ubiquitin-regulated gene expression of pro-

inflammatory cytokines such as IL1 β and IL6 may be involved in the induction of inflammation in the lysosomal damage response, in addition to the regulation at the stage of precursor processing. The present data on the regulation of gene expression by ubiquitin-mediated signal transduction may help elucidate the molecular mechanisms underlying diseases associated with lysosome dysfunction.

In this study, we demonstrated a pivotal role of the ubiquitin system in regulating gene expression in the lysosomal damage response, which extends its function beyond that in lysophagy initiation. We found that activation of the TAB-TAK1-IKK-NF- κ B pathway, potentially by the accumulation of K63 ubiquitin chains in damaged lysosomes, and the subsequent induction of gene expression are crucial for cell survival and cell-cell communication in cultured cells. Further analysis is required to determine the impact of this action on disease pathogenesis and the maintenance of cellular homeostasis in organisms.

While we mainly focused on the TAB-TAK1-IKK-NF- κ B pathway in this study, it should be noted that TAK1 also activates AMPK, subsequently promoting lysophagy¹⁵. AMPK activation by TAK1 may also contribute to NF- κ B activation or distinct gene regulation. We found that TBK1 is required for NF- κ B activation. In considering signal mediators such as AMPK and TBK1, it is postulated that cell homeostasis is cooperatively controlled by a set of regulators in the lysosomal damage response. A recent study demonstrated that linear ubiquitin chains formed by LUBAC and deubiquitinated by OTULIN activate NF- κ B in the lysosomal damage response⁵⁴. We have confirmed that LUBAC is essential for NF- κ B activation under our experimental conditions. It is well-documented that LUBAC-catalyzed M1 ubiquitin chains recruit IKK subunits and transduce the signaling to downstream mediators in the canonical pathway. Therefore, we hypothesized that K63 ubiquitin chains in damage lysosomes initially activate the TAB-TAK1 pathway and trigger LUBAC-mediated M1 ubiquitination. We further postulate that M1 ubiquitination subsequently recruits the IKK complex to damaged lysosomes. Consequently, activated TAK1 phosphorylates IKK subunits, leading to NF- κ B activation (**Fig. 6**). Apart from the lysosomal damage response, activated STING at the Golgi also leads to NF- κ B activation through LUBAC-mediated formation of linear ubiquitin chains⁶⁰. In addition, the IKK complex is recruited to damaged mitochondria, potentially through its association with accumulated ubiquitin chains, and activates NF- κ B⁶¹. Although the involvement of the TAB-TAK1 pathway was not examined in these studies, a common mechanism underlying NF- κ B activation induced by the accumulation of ubiquitin on damaged or stressed organelles may serve as a unifying determinant. Taken together with our previous findings that the accumulation of K63 ubiquitin chains in dysfunctional endosomes induces cytokine expression via the TAB-TAK1 pathway³², the present results suggest the existence of a universal response to the intracellular accumulation of K63 ubiquitin chains resulting from fluctuations in the cellular environment, including organelle stress. This universal response would entail the activation of NF- κ B and the modulation of downstream target gene expression for the regulation of cellular functions that maintain homeostasis.

Materials and methods

Cell culture

RPE-1 cells were cultured at 37°C under 5% CO₂ in Dulbecco's Modified Eagle Medium (DMEM) / Ham's F-12 (WAKO, 048-29785) supplemented with 10% fetal bovine serum (FBS, Nichirei, 175013), 1 mM sodium pyruvate (Thermo Fisher Scientific, 11360070), 1 $\times$ nonessential amino acids (Thermo Fisher Scientific, 11140050), and 1 $\times$ Penicillin-Streptomycin-Glutamine (Thermo Fisher Scientific, 10378016). HeLa cells were cultured at 37°C under 5% CO₂ in DMEM (Thermo Fisher Scientific, D5796) supplemented with 10% FBS, 1 mM sodium pyruvate, 1 $\times$ nonessential amino acids, and 1 $\times$ Penicillin-Streptomycin-Glutamine. For experiments using conditioned medium, RPE-1 cells were stimulated with culture medium from RPE-1 cells that had been treated with or without 1 mM L-

Leucyl-L-Leucine methyl ester (LLOMe, Cayman, 16008). At 30 min after treatment, the LLOMe was washed out and cells were further incubated in DMEM supplemented with 0.5% FBS, 1 mM sodium pyruvate, 1× nonessential amino acids, and 1× Penicillin–Streptomycin–Glutamine for 8 h. HeLa cells stably expressing siRNA-resistant FLAG-TAB2 WT, dNZF, and E685A were generated previously³².

Reagents and inhibitors

To induce lysosomal damage, cells were treated with 1 mM LLOMe, 200 μM Glycyl-L-phenylalanine 2-naphthylamide (GPN, Cayman, 14634), and DC661 (Cayman, 34899) for the indicated times. In the event of a longer incubation period, the LLOMe was washed out 2 h after treatment. The following inhibitors were used in this study: TAK-243 (10 μM, Active Biochem, A-1384), HS-276 (10 μM, Sigma-Aldrich, SML-3629), BX-795 (2 μM, Abcam, ab142016), and JNK-IN-8 (10 μM, Cayman, 18096).

Transcriptome analysis

Total RNA from RPE-1 cells was prepared using the Qiashredder kit (Qiagen, 79654), the RNeasy kit (Qiagen, 74104), and DNase (Qiagen, 79254). The quality of RNA samples was assessed using an Agilent 2100 bioanalyzer (Agilent Technologies); an RNA integrity number >9 was recorded for all samples. A total of 500 ng of each sample was processed with the Illumina Stranded mRNA Prep kit, and indexes were added with the IDT for Illumina RNA UD Indexes Set A (Illumina). The average library size was estimated using the Agilent 2100 Bioanalyzer, and libraries were quantified using a Qubit Fluorometer (Thermo Fisher Scientific). Samples were pooled into a single library and sequenced using the NextSeq 1000/2000 P2 Reagents (300 cycles, paired-end 150bp) on the Illumina NextSeq 1000 system. Sequence data were analyzed using the onboard DRAGEN RNA Pipeline Application (v.3.10.12).

Proteome analysis

RPE-1 cells were lysed in 5% SDS lysis buffer [5% SDS and 50 mM triethylammonium bicarbonate (TEAB, Thermo Fisher Scientific, 90114), followed by sonication with an ultrasonication probe. Cell lysates were reduced and alkylated with 4.5 mM dithiothreitol (DTT, Thermo Fisher Scientific, A39255) for 30 min at 55°C and 10 mM iodoacetamide (IAA, Thermo Fisher Scientific, A39271) for 15 min at room temperature (RT). Subsequently, 30 μg of cell lysates were loaded onto S-Trap micro columns (ProtiFi, C02- micro); trypsin and lysyl-endopeptidase (Lys-C) solution (1:10 w/w, Thermo Fisher Scientific, A41009) was added and incubated for 16 h at 37°C. The eluted peptides were dried in a vacuum concentrator and resuspended in 0.1% trifluoroacetic acid (TFA). Aliquots containing 500 ng of peptides from each sample were loaded onto a Vanquish Neo UHPLC system-connected Orbitrap Exploris 480 mass spectrometer (Thermo Fisher Scientific). The peptides were separated on an analytical column (C18, 1.7 μm particle size × 75 μm diameter × 600 mm, IonOpticks, AUR3-60075C18) heated at 55°C in a column oven (Sonation, PRSO-C2) with a constant flow rate of 250 nL/min. The peptides were eluted with a 0–40% acetonitrile gradient over 120 min. Peptide ionization was performed using the Nanospray Flex Ion Source (Thermo Fisher Scientific). The Orbitrap Exploris 480 mass spectrometer was operated in data-independent acquisition (DIA) mode using a full scan (*m/z* range of 380–985, nominal resolution of 60,000, and target value of 3×10^6 ions) followed by DIA-MS scans (fixed collision energy of 30%, isolation width of 10 *m/z* with an overlapping of 1 *m/z*, nominal resolution of 15,000, and target value of 2×10^6 ions).

DIA-MS data processing and visualization

The MS raw files were searched against the human UniProt reference proteome (Uniprot ID: UP000005640, reviewed, canonical, 20,563 entries) in library-free mode using DIA-NN software (version 1.9.1)⁶². The parameters that differed from the default settings were as follows: variable modifications, oxidation of methionine and acetylation of the peptide N-terminus; precursor *m/z* range, 380–985. Following processing with DIA-NN, the exported data were

subjected to imputation using Perseus software⁶³. The default setting was used, assuming that missing values represent low protein abundance. In this case, missing values were replaced by random numbers drawn from a normal distribution across the entire matrix. Gene enrichment tests and clustering were performed using the RNAseqChef software⁶⁴, and heatmaps were generated using the Morpheus platform (<https://software.broadinstitute.org/morpheus/>). The volcano plots, scatter plots, and bar graphs were visualized using GraphPad Prism software (version 8.1.0, GraphPad).

Cell lysis and immunoblotting

Cells were lysed with 2% SDS lysis buffer (2% SDS, 20 mM HEPES pH 7.5, 1 mM EDTA) containing a complete protease inhibitor cocktail (EDTA-free, Roche, 05056489001) and a phosphatase inhibitor cocktail (Roche, 4906845001). Cell lysates were boiled in 1× LDS NuPAGE sample buffer (Thermo Fisher Scientific, NP0008) for 10 min at 70°C, and then electrophoresed on 4–12% NuPAGE Bis-Tris gels (Thermo Fisher Scientific). Proteins were transferred to polyvinylidene difluoride membranes (Millipore, IPVH00010 or Pall, EH-2222). The membranes were incubated with 5% nonfat milk for 1 h at RT, followed by incubation for 2 h at RT with primary antibodies. The primary antibodies used for immunoblotting were anti-IL1 β mouse monoclonal (Cell Signaling Technology, 12242), anti-IL6 rabbit monoclonal (Cell Signaling Technology, 12153), anti-IRF1 rabbit monoclonal (Cell Signaling Technology, 8478), anti-NKX3.1 rabbit monoclonal (Cell Signaling Technology, 92998), anti-c-Fos rabbit monoclonal (Cell Signaling Technology, 2250), anti-c-Jun rabbit monoclonal (Cell Signaling Technology, 9165), HRP-conjugated anti- α -tubulin rabbit polyclonal (MBL, PM054-7), anti-K63 ubiquitin rabbit monoclonal (Millipore, 05-1308), anti-ubiquitin mouse monoclonal (Santa Cruz Biotechnology, sc-8017), anti-phospho-TAK1 (T184/T187) rabbit monoclonal (Cell Signaling Technology, 4508), anti-TAK1 rabbit polyclonal (Cell Signaling Technology, 4505), anti-TAB2 rabbit polyclonal (Abcam, ab222214), anti-TAB3 rabbit polyclonal (Abcam, ab85655), HRP-conjugated anti-FLAG mouse monoclonal (Sigma-Aldrich, A8592), anti-TFEB rabbit monoclonal (Cell Signaling Technology, 37785), anti-phospho-IKK α / β (S176/S180) rabbit monoclonal (Cell Signaling Technology, 2697), anti-IKK α rabbit polyclonal (Cell Signaling Technology, 2682), anti-IKK β rabbit monoclonal (Cell Signaling Technology, 8943), anti-phospho-I κ B α (S32) rabbit monoclonal (Cell Signaling Technology, 2859), anti-I κ B α rabbit monoclonal (Cell Signaling Technology, 4812), anti-phospho-JNK (T183/Y185) rabbit monoclonal (Cell Signaling Technology, 4668), anti-JNK rabbit polyclonal (Cell Signaling Technology, 9252), anti-phospho-p38 (T180/Y182) rabbit polyclonal (Cell Signaling Technology, 9211), anti-p38 rabbit monoclonal (Cell Signaling Technology, 8690), anti-phospho-ERK (T202/Y204) rabbit monoclonal (Cell Signaling Technology, 4370), anti-ERK rabbit monoclonal (Cell Signaling Technology, 4695), anti-cleaved caspase-3 rabbit polyclonal (Cell Signaling Technology, 9661), anti-phospho-STAT3 (Y705) rabbit monoclonal (Cell Signaling Technology, 9145), anti-STAT3 rabbit monoclonal (Cell Signaling Technology, 4904), anti-phospho-TBK1 (S172) rabbit monoclonal (Cell Signaling Technology, 5483), anti-TBK1 rabbit monoclonal (Abcam, ab40676), and anti-RNF31 rabbit monoclonal (Cell Signaling Technology, 99633) antibodies. The membranes were then incubated with secondary antibodies for 50 min at RT. The secondary antibodies used were HRP-conjugated goat anti-rabbit Ig (Promega, W4011) and anti-mouse Ig (Promega, W4021). The antibodies were diluted with Can Get Signal Solutions (TOYOBO, NKB-101). Chemiluminescence images were developed using the ECL Prime Western Blotting Detection Reagent (Cytiva, RPN2236) and acquired with a Fusion FX7 (Vilber Bio Imaging).

Phosphoproteome analysis

RPE-1 cells were treated with or without 1 mM LLOMe and 10 μ M HS-276 for the indicated times, after which they were lysed, reduced, and alkylated as described above (proteome analysis). Aliquots containing 200 μ g of lysate were loaded onto S-Trap mini columns (ProtiFi, C02-mini), followed by the addition of the trypsin solution (1:10 w/w, Cell Signaling Technology, 56296) and incubation for 16 h at 37°C. The eluted peptides were dried in a vacuum concentrator. Phosphopeptides were enriched using the High-Select Fe-NTA Phosphopeptide Enrichment Kit (Thermo Fisher Scientific, A32992) and dried in a vacuum concentrator. The resultant peptides

were cleaned using C18 spin tips (Thermo Fisher Scientific, 84850), dried in a vacuum concentrator, resuspended in 0.1% TFA, and loaded onto a Vanquish Neo UHPLC system-connected Orbitrap Exploris 480 mass spectrometer. The peptides were separated on an analytical column (C18, 1.7 μm particle size $\times$ 75 μm diameter $\times$ 250 mm, IonOpticks, AUR3-25075C18) heated at 55°C in a column oven with a constant flow rate of 250 nL/min. The peptides were eluted with a 0–40% acetonitrile gradient over 120 min. Peptide ionization was performed using the Nanospray Flex Ion Source. The Orbitrap Exploris 480 mass spectrometer was operated in data-dependent acquisition (DDA) mode utilizing a full scan (m/z range, 375–1500; nominal resolution, 60,000; target value, 3×10^6 ions) followed by MS/MS scans (fixed collision energy, 30%; isolation width, 1.6 m/z ; nominal resolution, 15,000, and target value, 1×10^5 ions). Precursor ions selected for fragmentation (charge state 2–6) were placed on a dynamic exclusion list for 20 s. A “1-s cycle” DDA method was used, whereby the most intense ions were selected every second for MS/MS fragmentation by higher-energy collisional dissociation.

DDA-MS data processing and visualization

The MS raw files were searched against the human UniProt reference proteome (Uniprot ID: UP000005640, reviewed, canonical, 20,563 entries) using the Sequest HT search program in Proteome Discoverer 3.1 (Thermo Fisher Scientific). Intensity-based non-label quantification was performed using the Precursor Ions Quantifier node in Proteome Discoverer 3.1. Volcano plots, scatter plots, and bar graphs were visualized using GraphPad Prism (version 8.1.0, GraphPad).

Immunofluorescence and confocal microscopy analysis

Cells were plated in 35-mm glass-bottomed dishes coated with poly-L-lysine (MatTek, P35-GC-0-10-C). They were fixed with 4% paraformaldehyde in PBS for 10 min at RT and permeabilized with 0.2% Triton X-100 in PBS for 5 min at RT. Alternatively, the cells were incubated with ice-cold methanol for 10 min on ice. Following incubation with 5% FBS and 0.1% Tween in PBS for 30 min at RT, cells were incubated with primary antibodies for 2 h at RT and then stained with secondary antibodies and DAPI (Thermo Fisher Scientific, D1306) for 1 h at RT. The following primary antibodies were used: anti-TAB2 mouse monoclonal (Santa Cruz Biotechnology, sc-398188), anti-LAMP1 rabbit monoclonal (Cell Signaling Technology, 9091), anti-K63 ubiquitin rabbit monoclonal (Millipore, 05-1308), anti-TAK1 mouse monoclonal (Santa Cruz Biotechnology, sc-7967), and anti-Galectin-3 rat monoclonal (Santa Cruz Biotechnology, sc-23938). The following secondary antibodies were purchased from Thermo Fisher Scientific: Alexa Fluor 488-conjugated anti-mouse (A-11029) and anti-rat (A-11006); and Alexa Fluor 594-conjugated anti-rabbit (A-11012). After staining, the cells were coverslipped (Matsunami, C015001) with SlowFade Gold (Thermo Fisher Scientific, S36936). Images were captured using ZEN 3.8 imaging software and LSM980 laser-scanning confocal microscopes equipped with a Plan-Apochromat 63 $\times$ /1.4NA oil lens (Carl Zeiss).

Small interfering RNA (siRNA) transfection

Cells were transfected with siRNAs using Lipofectamine RNAiMax (Thermo Fisher Scientific) at a final siRNA concentration of 30 nM. Cells were harvested and analyzed 72 h after transfection with siRNA. The siRNAs utilized pools of four different sequences (Horizon Discovery). The following siRNAs were used: TAB2 siRNAs: L-004771; TAB3 siRNAs: L-015572; TAK1 siRNAs: L-003790; IKK α siRNAs: L-003473; IKK β siRNAs: L-003503; IKK γ siRNAs: L-003763; and RNF31 siRNAs: L-021419.

RT-qPCR analysis

Total RNA was extracted using NucleoSpin RNA Plus (MACHEREY-NAGEL, 740984.250). cDNA was generated using the ReverTra Ace qPCR RT Master Mix with gDNA Remover kit (TOYOBO, FSQ-301). Quantification of mRNA was performed using a Light Cycler 480 (Roche) with THUNDERBIRD SYBR qPCR Mix (TOYOBO, QPS-201) as the detection reagent. The primer sets used for qRT-PCR were as follows: IL1 β forward: TACGATCACTGAAGTGCACGC, IL1 β reverse: CTTGTTGCTCCATATCTGTCCC; IL6 forward: TCATCACTGGTCTTTGGAGTTTG, IL6 reverse:

CAGCTCTGGCTTGTTCCTCAC; IRF1 forward: GCCATTCACACAGGCCGATAC, IRF1 reverse: TGCTCTGGTCTTTACCTCCTC; NKX3.1 forward: CTGGGAGACTTGGAGAAGCAC, NKX3.1 reverse: GGATAGCTGTTATACACGGAGACC; and GAPDH forward: AGAAGGTGGTGAAGCAGGCG, GAPDH reverse: CAAAGTGGTCGTTGAGGGCAATG.

Cell death assay

Cell death upon lysosomal damage was assessed by propidium iodide (PI) staining. Both cells attached to the culture dish and suspended in media were collected and stained with 50 µg/mL of PI (Thermo Fisher Scientific, P3566) for 15 min at RT. After washing excess PI, cells were resuspended in fresh PBS. Stained cells were loaded and counted on LSR- Fortessa (BD Biosciences).

Statistical analysis

Statistical analysis was performed using GraphPad Prism (version 8.1.0). All statistical information is provided in the figure legends. First, the sample distribution was assessed using the Shapiro–Wilk test. In the absence of formal testing, datasets with small sample sizes were assumed to have a normal distribution. The unpaired two-tailed Student’s t-test was used to determine statistical significance when comparing unpaired two independent groups with normal distribution and no significant difference in standard deviation (SD). For multiple comparisons involving more than two unpaired groups with normal distribution, an ordinary one-way ANOVA with Dunnett’s multiple comparison test was used. In all instances, statistical significance was evaluated with a 95% confidence interval, and a *P* value <0.05 was considered statistically significant.

Data availability

The RNA sequencing data have been deposited with links to BioProject accession number PRJDB19809 in the BioProject database of DNA Data Bank of Japan (DDBJ). The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository⁶⁵ with dataset identifiers PXD058072 (related to **Fig. 1B–E** and **S1A, B**), PXD058075 (related to **Fig. 4A, B**), and PXD058080 (related to **Fig. S4A–C**).

Supplementary figure legends

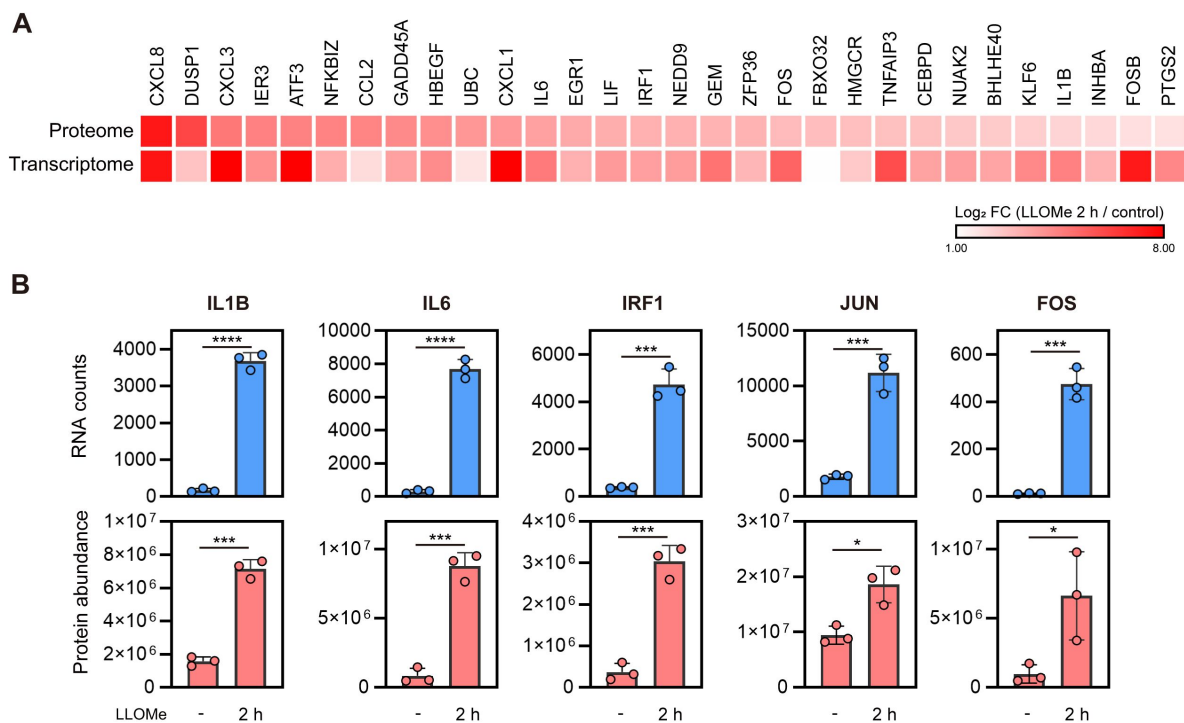


Figure S1

(A) Heatmap showing genes upregulated at both the protein and mRNA levels, as derived from **Fig. 1C** (top 30 in proteomic upregulation). The color scale represents Log₂ FC (LLOMe 2 h/control).

(B) RNA counts (top) and protein abundance (bottom) of representative genes, as measured by RNA sequencing and MS analysis, respectively. The individual values, mean, and SEM are shown. The mean $\pm$ SEM values were calculated from three biological replicates. * P < 0.05, *** P < 0.001, and **** P < 0.0001 (two-tailed Student's t -test).

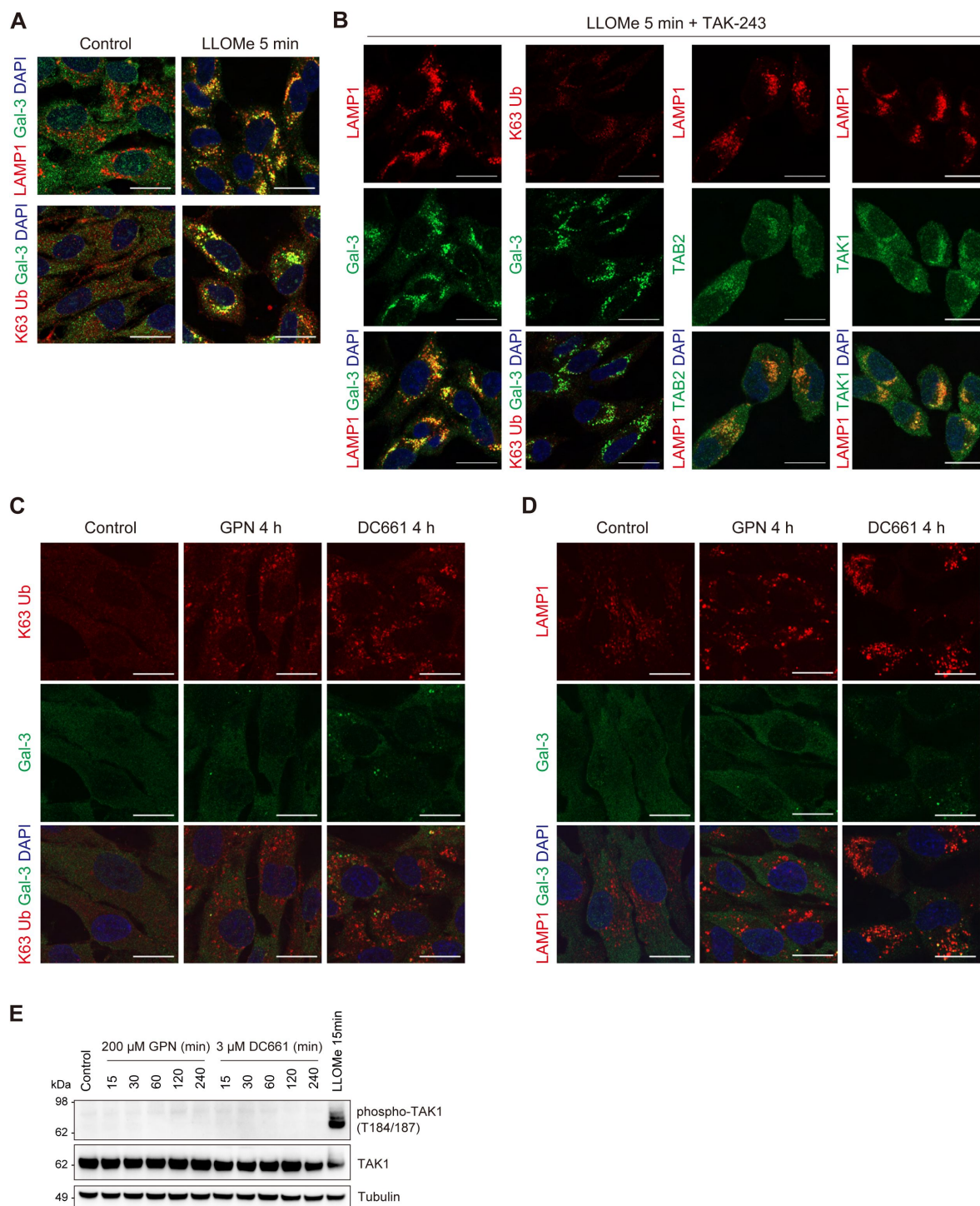


Figure S2

(A) RPE-1 cells treated with LLOMe for 5 min were immunostained with the indicated antibodies and DAPI. Scale bar, 20 μ m.

(B) RPE-1 cells pre-treated with TAK-243 for 15 min and treated with LLOMe for 5 min were immunostained with the indicated antibodies and DAPI. Scale bar, 20 μ m.

(C, D) RPE-1 cells treated with GPN (C) and DC661 (D) for 4 h were immunostained with the indicated antibodies and DAPI. Scale bar, 20 μ m.

(E) Total cell lysates from RPE-1 cells treated with GPN, DC661, and LLOMe for the indicated times were subjected to immunoblotting with the indicated antibodies.

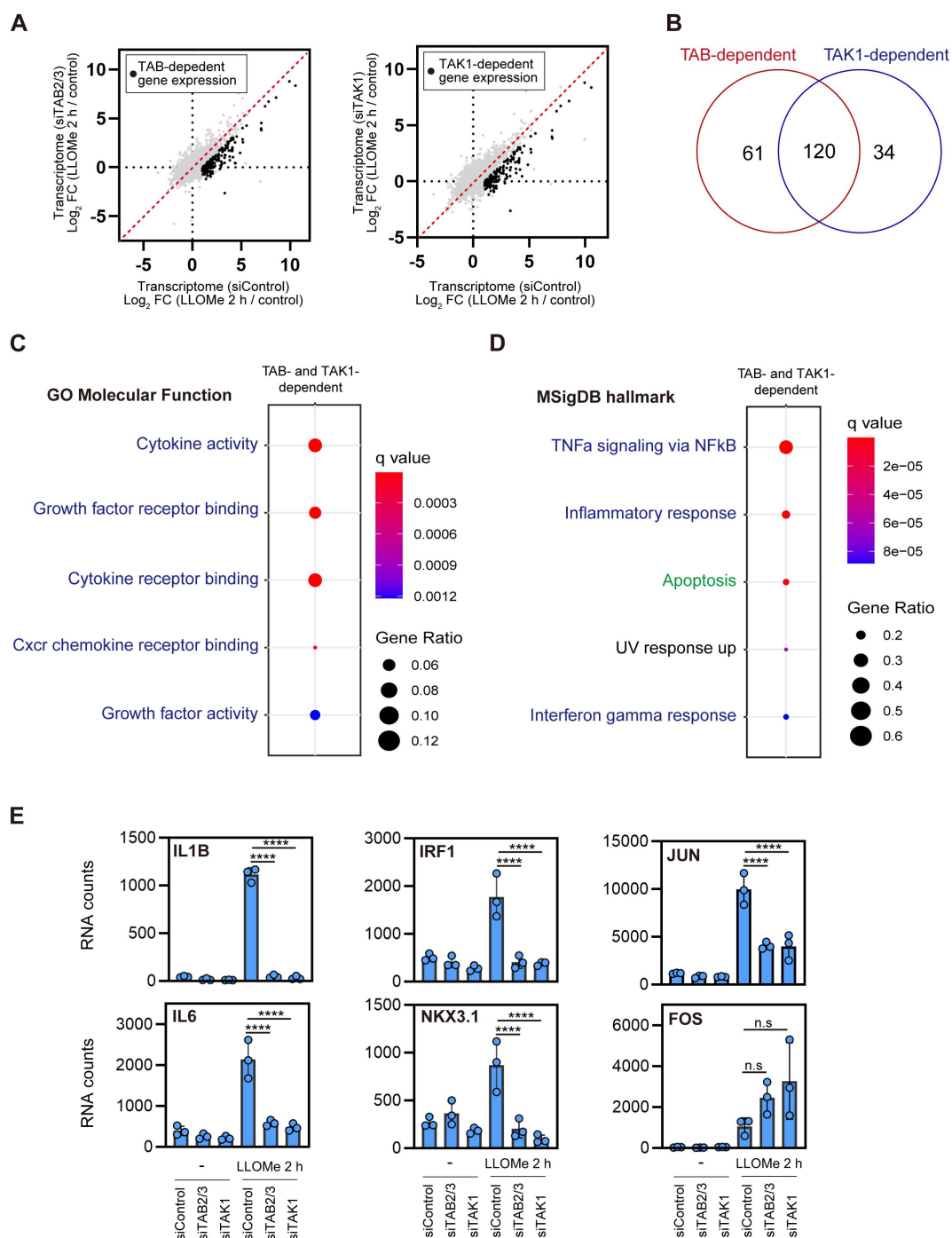


Figure S3

(A) Scatter plots showing the correlation between the mean Log₂ FC (LLOMe 2 h / control) in cells transfected with siControl (x axis) and siTAB2/3 (y axis) (left), and siControl (x axis) and siTAK1 (y axis) (right).

(B) Venn diagram showing the overlap of genes upregulated by LLOMe treatment in a TAB- and TAK1-dependent manner.

(C, D) The bubble plots show the outcomes of gene enrichment analyses based on GOMF (C) and MSigDB hallmark gene sets (D) for genes upregulated by LLOMe treatment in a TAB- and TAK1-dependent manner (top five categories). The color and size of the bubbles indicate the *q* value and gene ratio, respectively. The categories related to cytokine/growth factor and apoptosis are labeled in blue and green, respectively.

(E) RNA counts of representative genes, as measured by RNA sequencing. The individual values, mean, and SEM are presented. The mean $\pm$ SD values were calculated from three biological replicates. *****P* < 0.0001 (one-way ANOVA with Dunnett's test).

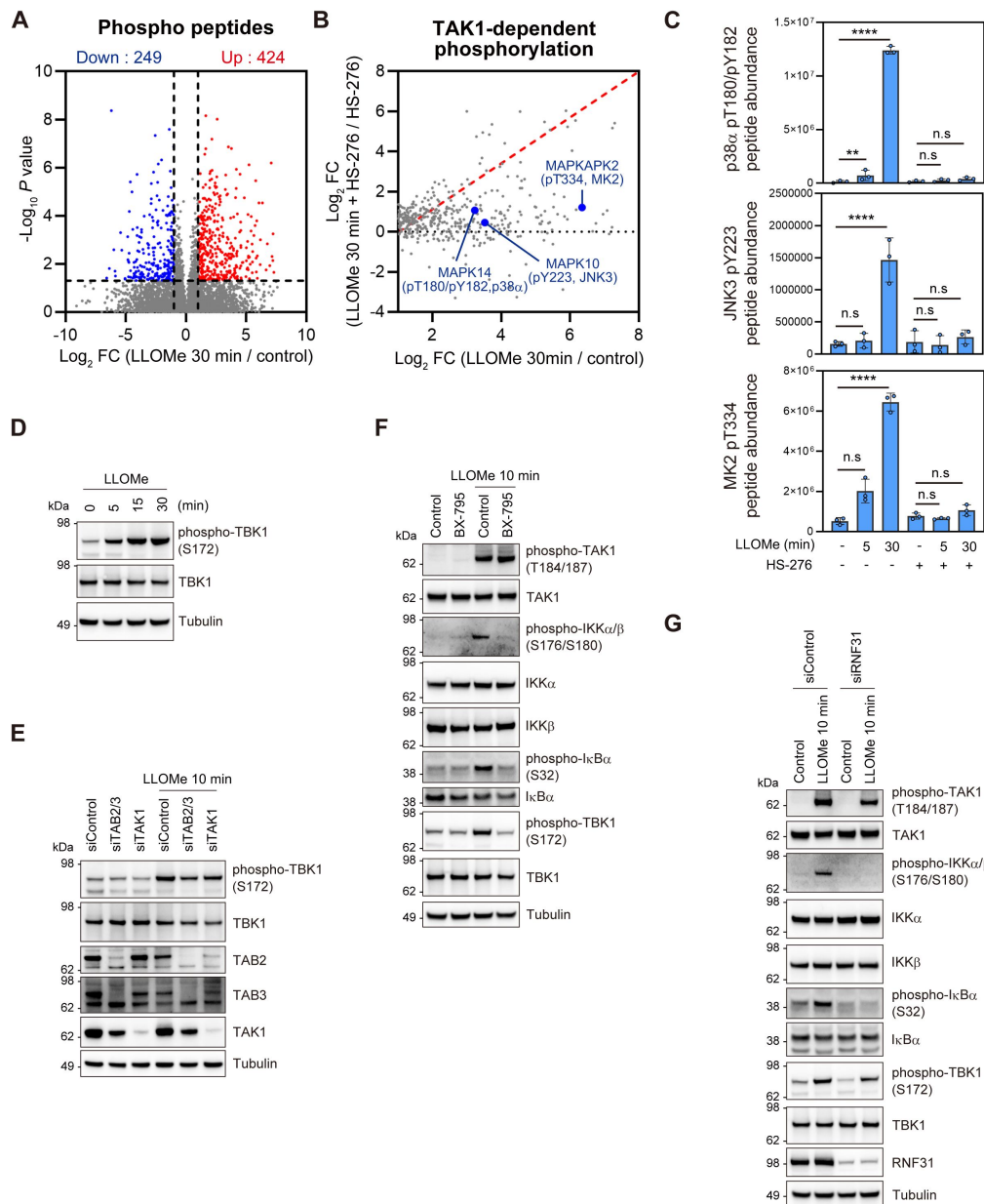


Figure S4

(A) The mean $\log_2 FC$ (LLOMe 30 min/control) and $-\log_{10} P$ value of the phosphopeptides are indicated on the x and y axes, respectively. Phosphopeptides significantly upregulated or downregulated are labeled in red and blue, respectively.

(B) Correlation between the mean $\log_2 FC$ (LLOMe 30 min/control) in cells treated with (y axis) or without HS-276 (x axis) is shown for phosphopeptides that were significantly upregulated 30 min after LLOMe treatment ($\log_2 FC > 1$ and $P < 0.05$).

(C) Phosphopeptides abundance measured by MS analysis. The individual values, mean, and SD of the mean of phosphopeptide abundance are shown. The mean $\pm$ SD values were calculated from three biological replicates. $**P < 0.01$ and $****P < 0.0001$ (one-way ANOVA with Dunnett's test).

(D) Total cell lysates from RPE-1 cells treated with LLOMe for the indicated times were subjected to immunoblotting with the indicated antibodies.

(E) Total cell lysates from RPE-1 cells transfected with the indicated siRNAs and treated with LLOMe for 10 min were subjected to immunoblotting with the indicated antibodies.

(F) Total cell lysates from RPE-1 cells pre-treated with BX-795 for 15 min and treated with LLOMe for 10 min were subjected to immunoblotting with the indicated antibodies.

(G) Total cell lysates from RPE-1 cells transfected with the indicated siRNAs and treated with LLOMe for 10 min were subjected to immunoblotting with the indicated antibodies.

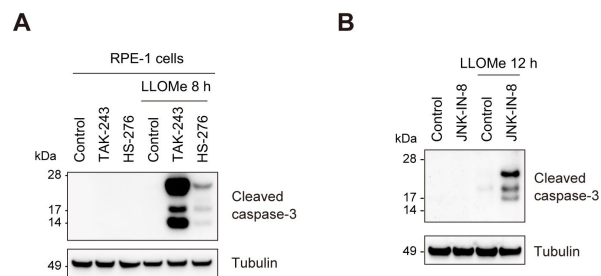


Figure S5

(A) Total cell lysates from RPE-1 cells pre-treated with TAK-243 or HS-276 for 15 min, treated with LLOMe for 2 h, and washed for 6 h were subjected to immunoblotting with the indicated antibodies.

(B) Total cell lysates from HeLa cells pre-treated with JNK-IN-8 for 15 min, treated with LLOMe for 2 h, and washed for 10 h were subjected to immunoblotting with the indicated antibodies.

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Additional information

Author contributions

The study was conceived by Akinori Endo, Keiji Tanaka, and Yukiko Yoshida. Transcriptome analyses were performed by Yasumasa Nishito. Proteome analyses were performed by Akinori Endo. Microscopy analyses were performed by Akinori Endo. Biochemical experiments were performed by Akinori Endo, Chikage Takahashi, Naoko Ishibashi, and Koji Yamano. Writing and editing were performed by Akinori Endo and Yukiko Yoshida.

Additional files

Table S1. Transcriptome analysis LLOMe 2 h [↗](#)

Table S2. Proteome analysis LLOMe 2 h [↗](#)

Table S3. Correlation of Transcriptome and Proteome LLOMe 2 h [↗](#)

Table S4. Transcriptome analysis RNAi LLOMe 2 h [↗](#)

Table S5. Proteome analysis LLOMe 30 min [↗](#)

Table S6. Phosphoproteome analysis with HS-276 [↗](#)

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Reviewer #1 (Public review):

Summary:

Lysosomal damage is commonly found in many diseases including normal aging and age-related disease. However, the transcriptional programs activated by lysosomal damage has not been thoroughly characterized. This study aims to investigate lysosome damage-induced major transcriptional responses and the underlying signaling basis. The authors have convincingly shown that lysosomal damage activates a ubiquitination-dependent signaling axis involving TAB, TAK1, and IKK, which culminate in the activation of NF- κ B and subsequent transcriptional upregulation of pro-inflammatory genes and pro-survival genes. Overall, the major aims of this study are successfully achieved.

Strengths:

This study is well-conceived and strictly executed, leading to clear and well-supported conclusions. Through unbiased transcriptomics and proteomics screens, the authors identifies NF- κ B as a major transcriptional program activated upon lysosome damage. TAK1 activation by lysosome damage-induced ubiquitination is found to be essential for NF- κ B activation and MAP kinase signaling. The transcriptional and proteomic changes are shown to be largely driven by TAK1 signaling. Finally, the TAK1-IKK signaling is shown to provide resistance to apoptosis during lysosomal damage response. The main signaling axis of this pathway has been convincingly demonstrated.

Overall, this study identifies major transcriptional responses following lysosomal damage through unbiased approaches. It is important to consider the impact of these pathways in disease settings where lysosomal integrity is compromised.

Comments on revisions:

The authors have adequately addressed all previous comments. I have no further recommendations.

<https://doi.org/10.7554/eLife.106901.2.sa3>

Reviewer #2 (Public review):

Summary:

Endo et al. investigate the novel role of ubiquitin response upon lysosomal damage in activating cellular signaling for cell survival. The authors provide a comprehensive transcriptome and proteome analysis of aging-related cells experiencing lysosomal damage, identifying transcription factors involved in transcriptome and proteome remodeling with a focus on the NF- κ B signaling pathway. They further characterized the K63-ubiquitin-TAB-TAK1-NF- κ B signaling axis in controlling gene expression, inflammatory responses, and apoptotic processes.

Strengths:

In the aging-related model, the authors provide a comprehensive transcriptome and characterize the K63-ubiquitin-TAB-TAK1-NF- κ B signaling axis. Through compelling experiments and advanced tools, they elucidate its critical role in controlling gene expression, inflammatory responses, and apoptotic processes.

Weaknesses:

The study lacks deeper connections with previous research, particularly:

- The established role of TAB-TAK1 in AMPK activation during lysosomal damage
- The potential significance of TBK1 in NF- κ B signaling pathways

Comments on revisions:

The authors have successfully addressed all the raised questions and the manuscript is now significantly improved.

<https://doi.org/10.7554/eLife.106901.2.sa2>

Reviewer #3 (Public review):

Summary:

The response to lysosomal damage is a fast-moving and timely field. Besides repair and degradation pathways, increasing interest has been focusing on damaged-induced signaling. The authors conducted both transcriptomics and proteomics to characterize the cellular response to lysosomal damage. They identify a signaling pathway leading to activation of NF κ B. Based on this and supported by Western blot and microscopy data, the authors nicely show that TAB2/3 and TAK1 are activated at damaged lysosomes and kick off the pathway to alter gene expression, which induces cytokines and protect from cell death. TAB2/3 activation is proposed to occur through K63 ubiquitin chain formation. Generally, this is a careful and well conducted study that nicely delineates the pathway under lysosomal stress. The "omics" data serves a valuable resource for the field. More work should be invested into how TAB2/3 are activated at the damaged lysosomes, also to increase novelty in light of previous reports.

Strengths:

Generally, this is a careful and well-conducted study that nicely delineates how the NF κ B pathway is activated under lysosomal stress and modulates cell behavior. The "omics" data serves as a valuable resource for the field.

Weaknesses:

While activation of TAB2/3 by K63-linked Ub chains is convincing, more work needs to be done on how they are recruited by distinct damage types to probe relevance for different pathophysiological conditions."

Comments on revisions:

The authors have addressed much of my criticism. Specifically, they have put (with new experiments) the data on the TAB2/3-TAK1 pathway in perspective to the previously reported LUBAC-mediated activation of NFkB. They also addressed the question about the significance of K63-linked chains for TAB2/3 activation with new complementation experiments (a K63-specific NZF mutant failed to rescue).

The third point (types of damage as triggers) raises more questions, though. The authors find that, in contrast to LLOMe, GPN or DC661-induced damage does not activate TAK1 (consistent with lower damage levels). However, the authors still observe K63 ubiquitylation. This goes along with their finding that TAB2 is recruited in the absence of any ubiquitylation (blocked by TAK-243). It argues that TAB2 is recruited by an unknown cue (that may be damage-specific) and then activated by K63. The authors need to clarify whether TAB2 is or is not recruited in the GPN/DC661 conditions (in which K63 occurs, but TAK1 is not activated). The point about the effects of other damage types was also raised by reviewer #1 and should be solved. The fact that TAB2 is recruited independently of K63 should also be visualized in the model. The manuscript will then be an important contribution to the field.

<https://doi.org/10.7554/eLife.106901.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

Lysosomal damage is commonly found in many diseases including normal aging and age-related disease. However, the transcriptional programs activated by lysosomal damage have not been thoroughly characterized. This study aimed to investigate lysosome damage-induced major transcriptional responses and the underlying signaling basis. The authors have convincingly shown that lysosomal damage activates a ubiquitination-dependent signaling axis involving TAB, TAK1, and IKK, which culminates in the activation of NF-kB and subsequent transcriptional upregulation of pro-inflammatory genes and pro-survival genes. Overall, the major aims of this study were successfully achieved.

Strengths:

This study is well-conceived and strictly executed, leading to clear and well-supported conclusions. Through unbiased transcriptomics and proteomics screens, the authors identified NF-kB as a major transcriptional program activated upon lysosome damage. TAK1 activation by lysosome damage-induced ubiquitination was found to be essential for NF-kB activation and MAP kinase signaling. The transcriptional and proteomic changes were shown to be largely driven by TAK1 signaling. Finally, the TAK1-IKK signaling was shown to provide resistance to apoptosis during lysosomal damage response. The main signaling axis of this pathway was convincingly demonstrated.

Weaknesses:

One weakness was the claim of K63-linked ubiquitination in lysosomal damage-induced NF- κ B activation. While it was clear that K63 ubiquitin chains were present on damaged lysosomes, no evidence was shown in the current study to demonstrate the specific requirement of K63 ubiquitin chains in the signaling axis being studied. Clarifying the roles of K63-linked versus other types of ubiquitin chains in lysosomal damage-induced NF- κ B activation may improve the mechanistic insights and overall impact of this study.

Another weakness was that the main conclusions of this study were all dependent on an artificial lysosomal damage agent. It will be beneficial to confirm key findings in other contexts involving lysosomal damage.

We would like to thank Reviewer #1 for the positive and constructive comments on our study. For a main concern regarding the molecular mechanism by which TAB proteins are activated in response to lysosomal damage, we have added the experimental results to support that the lysosomal accumulation of K63 ubiquitin chains serves as a trigger to activate the TAB-TAK1 pathway. We also investigated and discussed the role of LUBAC-mediated M1 ubiquitin chains in NF- κ B activation and the effects of other lysosomal-damaging compounds. Please see the response to “Reviewer #3 (Public review): Suggestions:”.

Reviewer #2 (Public review):

Summary:

Endo et al. investigate the novel role of ubiquitin response upon lysosomal damage in activating cellular signaling for cell survival. The authors provide a comprehensive transcriptome and proteome analysis of aging-related cells experiencing lysosomal damage, identifying transcription factors involved in transcriptome and proteome remodeling with a focus on the NF- κ B signaling pathway. They further characterized the K63-ubiquitin-TAB-TAK1-NF- κ B signaling axis in controlling gene expression, inflammatory responses, and apoptotic processes.

Strengths:

In the aging-related model, the authors provide a comprehensive transcriptome and characterize the K63-ubiquitin-TAB-TAK1-NF- κ B signaling axis. Through compelling experiments and advanced tools, they elucidate its critical role in controlling gene expression, inflammatory responses, and apoptotic processes.

Weaknesses:

The study lacks deeper connections with previous research, particularly:

- *The established role of TAB-TAK1 in AMPK activation during lysosomal damage*
- *The potential significance of TBK1 in NF- κ B signaling pathways*

We would like to thank Reviewer #2 for the helpful comments on our study. To achieve a more comprehensive understanding of the signaling pathways involved in the lysosomal damage response, we investigated additional related signal mediators, such as TBK1 and LUBAC. The citations related to AMPK have been incorporated.

Reviewer #3 (Public review):

Summary:

The response to lysosomal damage is a fast-moving and timely field. Besides repair and degradation pathways, increasing interest has been focusing on damaged-induced signaling. The authors conducted both transcriptomics and proteomics to characterize the cellular response to lysosomal damage. They identify a signaling pathway leading to activation of NFkappaB. Based on this and supported by Western blot and microscopy data, the authors nicely show that TAB2/3 and TAK1 are activated at damaged lysosomes and kick off the pathway to alter gene expression, which induces cytokines and protect from cell death. TAB2/3 activation is proposed to occur through K63 ubiquitin chain formation. Generally, this is a careful and well conducted study that nicely delineates the pathway under lysosomal stress. The "omics" data serves as a valuable resource for the field. More work should be invested into how TAB2/3 are activated at the damaged lysosomes, also to increase novelty in light of previous reports.

Strengths:

Generally, this is a careful and well-conducted study that nicely delineates the pathway under lysosomal stress. The "omics" data serves as a valuable resource for the field.

Weaknesses:

More work should be invested into how TAB2/3 are activated at the damaged lysosomes, also to increase novelty in light of previous reports. Moreover, different damage types should be tested to probe relevance for different pathophysiological conditions.

We would like to thank Reviewer #3 for the valuable comments on our study. We have added the experimental results to address two concerns raised by Reviewer #3. Please see the response to "Reviewer #3 (Public review): Suggestions:".

Suggestions:

(1) A recent paper claims that NFkappaB is activated by Otulin/M1 chains upon lysosome damage through TBK1 (PMID: 39744815). In contrast, Endo et al. nicely show that ubiquitylation is needed (shown by TAK-243) for NFkB activation but only have correlative data to link it specifically to K63 chains. On page 15, line 11, the authors even argue a "potential" involvement of K63. This point should be better dealt with. Can the authors specifically block K63 formation? K63R overexpression or swapping would be one way. Is the K63 ligase ITCH involved (PMID: 38503285) or any other NEDD4-like ligase? This could be compared to LUBAC inhibition. Also, the point needs to be dealt with more controversially in the discussion as these are alternative claims (M1 vs K63, TAB vs TBK1).

It is well-characterized that the NZF domain of TAB proteins preferentially associates with K63-linked ubiquitin chains. Therefore, we performed the add-back experiment using siRNA-resistant TAB2 WT and mutants incapable of binding to K63-linked ubiquitin chains, dNZF and E685A, to elucidate the requirement of K63 ubiquitin chains for TAK1 activation. We investigated whether the add-back of TAB2 mutants rescues the activation of TAK1 in TAB2-depleted cells (Fig. 2E). TAB2 WT, but not dNZF and E685A, rescued TAK1 activation in response to LLOMe, suggesting that the specific interaction of TAB proteins and K63 ubiquitin chains is a key mechanism to activate TAK1. We also found that the treatment of an E1 inhibitor TAK-243 effectively prevented the lysosomal accumulation of K63 ubiquitin chains, but TAB2 was recruited to damaged lysosomes (Fig. S2B). This suggests that the recruitment of TAB proteins to damaged lysosomes is independent of the association with K63 ubiquitin chains. Collectively, it is postulated that TAB proteins require interaction with K63 ubiquitin chains for TAK1 activation, but not for recruitment to damaged lysosomes. We have added the sentences (p9, lines 7-20, and p10, lines 8-10).

Next, we confirmed that LUBAC functions are essential for NF- κ B activation in the lysosomal damage response. RNF31/HOIP is a component of LUBAC that catalyzes M1 ubiquitination. The depletion of RNF31 showed no significant effects on TAK1 activation, but abolished IKK activation (Fig. S4G). It is well-characterized that LUBAC-mediated M1 ubiquitin chains recruit IKK subunits and transduce the signaling to downstream in the canonical pathway. We assume that K63 ubiquitin chains in damaged lysosomes initially activate TAB-TAK1 and trigger LUBAC-mediated M1 ubiquitination, and subsequently, M1 ubiquitination functions to recruit the IKK complex. Consequently, activated TAK1 phosphorylates IKK subunits in damaged lysosomes, leading to NF- κ B activation. We also examined whether TBK1 is involved in the activation of NF- κ B. TBK1 was phosphorylated upon LLOMe, and depletion of TAB and TAK1 resulted in a slight reduction of TBK1 phosphorylation (Fig. S4D, E). The treatment of a TBK1 inhibitor BX-795 exhibited no or little effects on TAK1 activation, but abolished phosphorylation of IKK and I κ Ba (Fig. S4F). These suggest that TBK1 is required for the activation of NF- κ B. We have added the sentences (p13, line 13-p14, line 10).

As mentioned by Reviewer #3, it is important to identify the E3 ligase responsible for K63 ubiquitination in the lysosomal damage response. We have been aiming to identify such E3 ligase(s). However, depletions of ITCH and other E3 ligases that have been tested exhibited no or little effects on K63 ubiquitination and TAK1 activation. We would like to explore E3 ligase(s) in future study.

(2) It would be interesting to know what the trigger is that induces the pathway. Lipid perturbation by LLOMe is a good model, but does activation also occur with GPN (osmotic swelling) or lipid peroxidation (oxidative stress) that may be more broadly relevant in a pathophysiological way? Moreover, what damage threshold is needed? Does loss of protons suffice? Can activation be induced with a Ca²⁺ agonist in the absence of damage?

To further clarify the initial trigger that induces TAB-TAK1 activation coupled with lysosomal damage, we examined other damage sources, GPN and DC661, which induce hyperosmotic stress and lipid peroxidation in lysosomes, respectively, thereby resulting in lysosomal membrane damage. Under our experimental conditions, the treatment of these compounds did not result in significant accumulation of Gal-3, indicating a reduced level of lysosomal membrane permeabilization compared with LLOMe (Fig. S2C, D), and no or little TAK1 activation was observed (Fig. S2E). TAB proteins require their association with K63 ubiquitin chains for TAK1 activation. It is therefore postulated that the severe lysosomal membrane permeabilization that triggers the formation and cytosolic exposure of K63 ubiquitin chains may be a determinant of TAB-TAK1 activation. In our future work, we would like to examine broad stimulation of lysosomal damage and further elucidate the initial mechanism of TAB-TAK1 activation. We have added the sentences (p9, line 21-p10, line 7).

(3) The authors nicely define JNK and p38 activation. This should be emphasized more, possibly also in the abstract, as it may contribute to the claim of increased survival fitness.

We further tested whether the inhibition of JNK affects the anti-apoptotic effect (Fig. S5B). The inhibition of JNK resulted in an increase in the cleaved caspase-3. This suggests that the anti-apoptotic action in the lysosomal damage response requires JNK as well as IKK. We have added the sentences in results to emphasize the pivotal role of stress-induced MAPKs (p15, lines 7-11).

Reviewer #1 (Recommendations for the authors):

(1) Although the ubiquitination-TAB-TAK1-IKK axis was previously characterized in other contexts, specific evidence supporting lysosomal recruitment of these components by ubiquitination during lysosome damage would be beneficial.

We found that the treatment of an E1 inhibitor TAK-243 abolished the lysosomal accumulation of K63 ubiquitin chains, but TAB2 and TAK1 were recruited to damaged lysosomes (Fig. S2B). This suggests that the recruitment of TAB proteins to damaged lysosomes is independent of the association with K63-linked ubiquitin chains. Next, we investigated whether the add-back of TAB2 mutants incapable of binding K63 ubiquitin chains rescues the activation of TAK1 in TAB2-depleted cells (Fig. 2E). K63 ubiquitin binding of TAB2 was essential for TAK1 activation in response to LLOMe. Taken together, it is suggested that TAB proteins require their interaction with K63 ubiquitin chains for TAK1 activation, but not for recruitment to damaged lysosomes. We have added the sentences (p9, lines 7-20, and p10, lines 8-10). Please also see the response to “Reviewer #3 (Public review): Suggestions:”.

(2) The activation of p38 and JNK by lysosomal damage does not fit well into the main conclusions of the paper, since IKK knockdown was sufficient to block cellular resistance to apoptosis (caspase cleavage in Fig. 5f). Are p38 and JNK also important for cell survival during lysosomal damage?

We found that the inhibition of JNK resulted in an increase in the cleaved caspase-3, suggesting that the anti-apoptotic action in the lysosomal damage response requires both IKK and JNK (Fig. S5B). We have added the sentences (p15, lines 7-11).

(3) Cell death tests are recommended to support the conclusions related to apoptosis.

As suggested by Reviewer #1, we performed the cell death assay using propidium iodide (PI) and confirmed that HeLa cells co-treated with LLOMe and TAK-243 or HS-276 exhibited increased cell death (Fig. 5E). This indicates a direct correlation between the degree of caspase-3 cleavage and cell death, possibly apoptosis.

(4) Page 8, line 19-21, gal3 is not exposed upon lysosomal damage. It is recruited from the cytosol by the exposed beta-galactoside-containing glycans on lysosomal membrane proteins.

We have corrected the corresponding sentence (p7, lines 17-20).

(5) Carefully checking grammar throughout the text is recommended. Below are a few examples:

a) Page 4, line 10, remove "that".

b) "K63 ubiquitin" shall be replaced with "K63 ubiquitination" or "K63 ubiquitin chains".

c) Page 8, line 9, "remain" should be "remains".

We have carefully checked the revised manuscript.

Reviewer #2 (Recommendations for the authors):

Despite the novelty and significance of these findings in advancing the field, several technical and experimental limitations require further clarification:

We have responded to each comment. Please see below.

The manuscript should introduce or discuss previous research showing that TAB-TAK1 facilitates AMPK activation during lysosomal damage and TAK1's increased association with damaged lysosomes (PMID: 31995728).

We have added the reference (PMID: 31995728) and the sentences (p17, lines 15-20).

Figure 2A: The differential LAMP1 staining intensity between control and LLOMe-treated cells needs explanation. The weaker LAMP1 signal in control and puncta changes, especially during 5-minute LLOMe treatment, require detailed clarification

We have added the explanation (p8, lines 17-21).

Recent literature (PMID: 34585663) reports TBK1 activation during lysosomal damage. The authors should investigate or discuss whether TBK1 potentially contributes to NF- κ B signaling in this context.

We experimentally investigated whether TBK1 is involved in the TAB-TAK1 pathway. We confirmed that TBK1 was activated upon LLOMe (Fig. S4D). Depletions of TAB and TAK1 exhibited a modest decrease in TBK1 phosphorylation (Fig. S4E). The inhibition of TBK1 by BX-795 did not affect TAK1 activation, but abolished phosphorylation of IKK and I κ Ba (Fig. S4F). This suggests that TBK1 is required for NF- κ B activation. We have added the reference (PMID: 34585663) and the sentences (p13, lines 13-21, p14, lines 8-10, and p18, lines 15-20).

The introduction of lysosomal damage response lacks comprehensive mechanistic information. For example, while ESCRT is discussed, other critical mechanisms such as lipid transfer and stress granule formation in lysosomal repair should be incorporated. Moreover, mTOR and AMPK signaling pathways undergo significant changes upon lysosomal damage.

We have added the sentences (p3, lines 16-18, and p3, line 21-p4, line 1).

The statement "lysosomal permeabilization causes the dissociation of mTORC1 from lysosomes" should explicitly reference PMID: 29625033.

We have added the suggested reference (PMID: 29625033, p4, line 19).

The claim that "The elimination of damaged lysosomes through lysophagy requires a period of more than half a day" needs a specific publication citation.

We have added the reference (PMID: 23921551) to claim the time-scale of lysosomal clearance (p4, line 21).

Figure 1G: The label "WO after 2h" lacks explanation in the figure legend and requires detailed interpretation.

To simplify the figures, we have deleted the label "WO after 2 h" (Fig. 1G, 3F, 5D, F-J, S4G, S5A). Instead, we have added the explanation in the figure legends (Fig. 1G).

Reviewer #3 (Recommendations for the authors):

(1) page 8, line 13: it is recommended to phrase colocalisation "at" damaged lysosomes rather than "in" damaged lysosomes as the resolution does not allow the claim of influx

| *into lysosomes.*

We have corrected the word (p8, line 17).

| *(2) page 11, line 22: why is "whereas" used to link two events driven by the same mechanism.*

We have corrected the word (p13, line 8).

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